

PROGRAMMING FUNDAMENTAL ASSIGNMENT

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Question 1: What is a variable in C/C++ ?

A variable is named memory location used to store data that can change during program execution .

Question 2 : Why do we use variables in programs ?

Variable are used to store data in program. They help us save ,access and modify values whenever needed during program execution.

Example : int marks =20 ;

Here marks stores student marks.

Question 3: What is the difference between a variable name and its value?

A variable name is the identifier used to refer to memory location ,while a variable value is the actual data stored in that location.

Example : `int marks = 40;`

- VARIABLE NAME : marks
- VARIABLE VALUE : 40

Question 4: Which of the following is a valid variable name?

- a) 2num
- b) total_marks
- c) for
- d) my marks

The correct answer is b) total_marks

Reason:

- 2num is invalid because variable names cannot start with a number.
- for is invalid because it is a reserved keyword.
- my marks is invalid because spaces are not allowed.
- total_marks is a valid variable name.

Question 5: Why is Age different from age?

C/C++ is a case-sensitive language, meaning uppercase and lowercase letters are treated

differently.

Example: `int Age = 20;`

`int age = 18;`

Here, Age and age are considered two different variables.

Question 6: Can a variable name contain spaces? Why?

No, a variable name cannot contain spaces because the compiler treats spaces as separators.

- Wrong Example:
 - `int my age = 18;`
- Correct Example:
 - `int my_age = 18;`

Question 7: Can a variable name start with an underscore (_) ?

Yes, a variable name can start with an underscore (_).

Example:

- `int _marks = 60;`

Question 8: What happens if two variables have the same name in the same scope?

The compiler gives an error because two variables cannot have the same name in the same scope.

Example:

```
int age = 18;
```

```
int age = 20;
```

This will generate an error.

Question 9: Write three meaningful variable names for storing student information.

- studentName
- studentAge
- studentMarks

These variable names are meaningful because they clearly describe the stored information.

Question 10: Which naming convention is better and why: studentMarks or sm?

studentMarks is better because it is meaningful, readable, and easy to understand.

sm is not clear and may confuse programmers about what data it stores.

Question 11: What is a data type?

A data type defines the type of data a variable can store in a program.

Example:

- int stores integers
- float stores decimal numbers
- char stores characters

```
int age = 13;  
float marks = 80.9;  
char grade = 'A';
```

Question 12: Which data type is used to store whole numbers?

The int data type is used to store whole numbers.

Example:

```
int age = 15;  
Here, 15 is a whole number stored using int.
```

Question 13: Which data type is used to store decimal values?

The float or double data type is used to store decimal values.

Example:

```
float price = 60.99;
```

Question 14: What is the difference between float and double?

float stores decimal numbers with less precision, while double stores decimal numbers with higher precision.

Example:

```
float num1 = 3.14;  
double num2 = 3.1415926535;  
double can store more digits after the decimal point.
```

Question 15: Which data type stores a single character?

The char data type is used to store a single character.

Example:

```
char grade = 'A';
```

Question 16: Which data type stores True/False values?

The bool data type is used to store True/False values.

Example:

```
bool isPassed = true;
```

Question 17: What data type is suitable for storing a person's name?

The string data type is suitable for storing a person's name.

Example:

```
string name = "kasaf";
```

Question 18: Predict the output:

```
int age = 20;  
cout << age;
```

Output:
20

Explanation:

The value stored in age variable is 18, so c out displays 18.

Question 19: Identify the data type: 25, 3.14, 'A', true.

Value	Data Type
-------	-----------

25	int
3.14	float/double
'A'	char
true	

Question 20: What happens if you store a decimal number in an int variable?

The decimal part gets removed automatically because int stores only whole numbers.

Example:

```
int num = 6.72;  
cout << num;
```

Output:

6

The decimal part (.72) is removed.

Question 21: What is the purpose of Cin?

Cin is used to take input from the user in C++.

Example:

```
int age;  
Cin >> age;  
Here, the user enters a value for age.
```


Question 22: What is the purpose of Cout?

Cout is used to display output on the screen in C++.

Example:

```
Cout << "Hello";
```

This will display Hello on the screen.

Question 23: Which symbol is used with Cin?

The extraction operator (>>) is used with Cin.

Example:

```
cin >> age;
```

Question 24: Which symbol is used with c out?

The insertion operator (<<) is used with c out.

Example:

```
cout << age;
```

Question 25: Explain:

```
c in >> age;
```

This statement takes input from the user and stores it

in the variable age.

Explanation:

- `c in` → takes input
- `>>` → extraction operator
- `age` → variable where input is stored
-

Example:

If user enters 18, then `age = 18`.

Question 26: Explain:

`c out << age;`

This statement displays the value stored in the variable age on the screen.

Explanation:

- `cout` → displays output
- `<<` → insertion operator
- `age` → variable whose value will be printed

Example:

If `age = 18`, output will be:

18

Question 27: What is the purpose of `endl`?

`endl` is used to move the cursor to the next line.

Example:

```
cout << "Hello" << endl;
```

```
cout << "World";
```

Output:

Hello

World

Question 28: Write a statement to input a student's marks.

```
int marks;
```

```
cin >> marks;
```

This statement takes marks as input from the user.

Question 29: Write a statement to display 'Welcome to C++'.

```
cout << "Welcome to C++";
```

Question 30: Predict the output:

```
cout << "Programming";
```

Output:

Programming

Explanation:

Cout prints the text written inside double quotation marks.

Question 31: What is an operator?

An operator is a symbol used to perform operations on variables and values.

Examples of Operators:

- + → Addition
- - → Subtraction
- * → Multiplication
- / → Division

Example:

```
int sum = 10 + 5;
```

Here, + is an operator.

Question 32: Which operator is used for addition?

The + (plus) operator is used for addition.

Example:

```
int sum = 4+6;
```

```
cout << sum;
```

Output:

10

Question 33: What is the result of $10 \% 3$?

The result is 1.

Explanation:

The $\%$ (modulus) operator gives the remainder after division.

$$10 \% 3 = 1$$

Because:

$$10 \div 3 = 3 \text{ remainder } 1$$

Question 34: What is the result of $8 \% 2$?

The result is 0.

Explanation:

$$8 \% 2 = 0$$

Because:

8 is completely divisible by 2, so remainder is 0.

Question 35: Which operator checks equality?

The $==$ operator is used to check equality.

Example:

if ($a == b$)

This checks whether a is equal to b.

Question 36: What is the difference between $=$ and

== ?

Operator	Purpose
=	Assignment operator
==	Equality operator

Example:

int a = 10;
= assigns value 10 to a.

if (a == 10)
== checks if a is equal to 10.

Question 37: What is the result of $5 > 3$?

The result is True.

Explanation:

Since 5 is greater than 3, the condition becomes true.
 $5 > 3 = \text{true}$

Question 38: What does && represent?

&& represents the Logical AND operator.
It returns true only when both conditions are true.

Example:

if (age > 18 && marks > 50)
Both conditions must be true.

Question 39: What does || represent?

|| represents the Logical OR operator.
It returns true if at least one condition is true.

Example:

if (age > 18 || marks > 50)
Only one condition needs to be true.

Question 40: What does ! represent?

! represents the Logical NOT operator.
It reverses the result of a condition.

Example:

!(5 > 3)
Since 5 > 3 is true, applying ! makes it false.
Result = false

Question 41: Write a program to print "Hello World"
Problem Statement.

Write a program to display "Hello World" on the screen.

Algorithm

- Start the program
- Display "Hello World" using cout
- End the program

C++ Program

```
#include <iostream>
using namespace std;
```

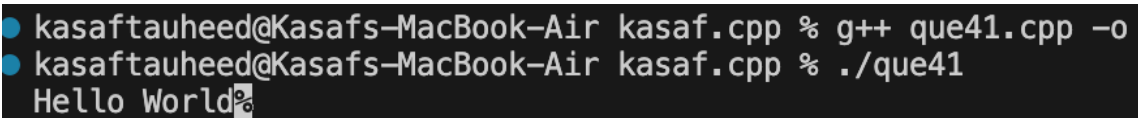
```
int main()
{
    cout << "Hello World";

    return 0;
}
```

Explanation

- `#include <iostream>` → Used for input/output operations
- `using namespace std;` → Allows use of cout without `std::`
- `cout` → Displays output on screen
- `return 0;` → Ends the program successfully

Output:



```
kasaftauheed@Kasafs-MacBook-Air kasaf.cpp % g++ que41.cpp -o
kasaftauheed@Kasafs-MacBook-Air kasaf.cpp % ./que41
Hello World%
```

Question 42: Write a program to input two numbers

and display their sum
Problem Statement.

Write a program to take two numbers from the user and display their sum.

Algorithm

- Start the program
- Declare two variables
- Take input from user
- Add both numbers
- Display the result
- End the program
-

C++ Program

```
#include <iostream>
using namespace std;
```

```
int main()
{
    int num1, num2, sum;

    cout << "Enter first number: ";
    cin >> num1;

    cout << "Enter second number: ";
    cin >> num2;

    sum = num1 + num2;

    cout << "Sum = " << sum;
```

```
    return 0;  
}
```

Explanation

- cin takes input from the user
- num1 + num2 adds both numbers
- Result is stored in sum
- cout displays the result
-

Output :

```
kasaf@kasafs-MacBook-Air kasaf.cpp % g++ que42.cpp -o  
kasaf@kasafs-MacBook-Air kasaf.cpp % ./que42  
Enter first number: 20  
Enter second number: 30  
Sum = 50%
```

Question 43: Write a program to calculate the area of a rectangle

Problem Statement.

Write a program to calculate the area of a rectangle.

Formula:

Area = Length × Breadth

Algorithm

- Start the program
- Declare variables for length and breadth
- Take input from user
- Calculate area
- Display area

- End the program

C Program

```
#include <stdio.h>
```

```
int main()
```

```
{
```

```
    float length, breadth, area;
```

```
    printf("Enter length: ");
```

```
    scanf("%f", &length);
```

```
    printf("Enter breadth: ");
```

```
    scanf("%f", &breadth);
```

```
    area = length * breadth;
```

```
    printf("Area of Rectangle = %.2f", area);
```

```
    return 0;
```

```
}Explanation
```

- scanf() takes input
- length * breadth calculates area
- printf() displays output
- %.2f shows decimal values up to 2 places

Sample Run:

```

kasaftauheed@Kasafs-MacBook-Air c practice.1 % gcc que43.c -o
kasaftauheed@Kasafs-MacBook-Air c practice.1 % ./que43
Enter length: 12
Enter breadth: 18
Area of Rectangle = 216.00%
kasaftauheed@Kasafs-MacBook-Air c practice.1 %

```

Question 44: What formula is used to calculate Simple Interest?

The formula used to calculate Simple Interest (SI) is:

$$SI = \frac{P \times R \times T}{100}$$

Where:

- P = Principal Amount
- R = Rate of Interest
- T = Time
- SI = Simple Interest
-

Example :

If:

P = 1000, R = 5, T = 2

Then:

$SI = (1000 \times 5 \times 2) / 100 = 100$

Question 45: Write a program to calculate Simple Interest

Problem Statement

Write a program to calculate Simple Interest using principal, rate, and time.

Algorithm

- Start the program
- Declare variables
- Take principal, rate, and time as input
- Apply formula
- Display result
- End the program
-

C Program

```
#include <stdio.h>
```

```
int main()
```

```
{
```

```
    float principal, rate, time, SI;
```

```
    printf("Enter Principal Amount: ");  
    scanf("%f", &principal);
```

```
    printf("Enter Rate of Interest: ");  
    scanf("%f", &rate);
```

```
    printf("Enter Time: ");  
    scanf("%f", &time);
```

```
    SI = (principal * rate * time) / 100;
```

```
    printf("Simple Interest = %.2f", SI);
```

```
    return 0;
```

```
}
```

Explanation

- User enters principal amount, rate, and time
- Formula is applied to calculate simple interest
- Result is displayed using printf()

Sample Run:

```
kasaftauheed@Kasafs-MacBook-Air c practice.1 % gcc assignment2.c
kasaftauheed@Kasafs-MacBook-Air c practice.1 % ./assignment2
Enter Principal Amount: 200
Enter Rate of Interest: 2
Enter Time: 4
Simple Interest = 16.00%
```

Question 46: Why is a temporary variable used while swapping two numbers?

A temporary variable is used to safely store one value while swapping two numbers so that the original value is not lost.

Example :

Suppose:

a = 1

b = 2

If we directly do:

a = b;

b = a;

Both values become 2, which is wrong.

So we use a temporary variable:

temp = a;

a = b;

b = temp;

Now values are swapped correctly.

Question 47: Swap the values: a = 10, b = 20
Problem Statement.

Write a program to swap two numbers using a temporary variable.

Algorithm

- Start the program
- Declare variables a, b, and temp
- Store value of a in temp
- Assign value of b to a
- Assign value of temp to b
- Display swapped values
- End the program
-

C++ Program

```
#include <iostream>
```

```
using namespace std;
```

```
int main()
```

```
{
```

```
    int a = 10, b = 20, temp;
```

```
    cout << "Before Swapping:" << endl;
```

```
    cout << "a = " << a << endl;
```

```
    cout << "b = " << b << endl;
```

```

temp = a;
a = b;
b = temp;

cout << "After Swapping:" << endl;
cout << "a = " << a << endl;
cout << "b = " << b << endl;

return 0;
}

```

Explanation

- temp = a stores value of a
- a = b copies value of b into a
- b = temp restores original value of a into b
- This swaps both numbers successfully

Output :



```

kasaf@kasafs-MacBook-Air kasaf.cpp % g++ que47.cpp -o o
kasaf@kasafs-MacBook-Air kasaf.cpp % ./que47
Before Swapping:
a = 10
b = 20
After Swapping:
a = 20
b = 10

```

Question 48: What formula is used to convert Celsius into Fahrenheit?

The formula used to convert Celsius into Fahrenheit is:

$$F = \left(\frac{9}{5} \times C\right) + 32$$

Where:

- F = Fahrenheit
- C = Celsius
-

Example :

If Celsius = 0°C

Then:

$$F = (9/5 \times 0) + 32$$

$$F = 32^{\circ}\text{F}$$

Question 49: What is the Fahrenheit value of 0°C?

Using the formula:

$$F = \left(\frac{9}{5} \times C\right) + 32$$

When:

$$C = 0^{\circ}\text{C}$$

Calculation:

$$F = (9/5 \times 0) + 32$$

$$F = 32^{\circ}\text{F}$$

Final Answer:

The Fahrenheit value of 0°C is 32°F.

Question 50: Why can integer division give incorrect results in the Celsius-to-Fahrenheit program?

Integer division can give incorrect results because integers remove the decimal part.

For example:

5 / 9

This becomes:

0

instead of:

0.555...

So, if integer division is used in the Celsius-to-Fahrenheit formula, the answer may become wrong.

Correct Way

Use float or double for accurate decimal calculations.

Example Program (C)

```
#include <stdio.h>
```

```
int main()
```

```
{
```

```
    float celsius, fahrenheit;
```

```
    printf("Enter temperature in Celsius: ");
```

```
    scanf("%f", &celsius);
```

```
    fahrenheit = (9.0 / 5.0 * celsius) + 32;
```

```
    printf("Temperature in Fahrenheit = %.2f",  
    fahrenheit);
```

```
    return 0;
```

}

Sample Run

```
• kasaftauheed@Kasafs-MacBook-Air c practice.1 % gcc assignment1
• kasaftauheed@Kasafs-MacBook-Air c practice.1 % ./assignment1
Enter temperature in Celsius: 30
Temperature in Fahrenheit = 86.00%
```